REFINING PH COVERINGS OF AFFINES BY STANDARD PH COVERINGS AFTER SHRINKING THE TARGET

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [2553, 2563] and proof(s) [[2565, 2623]]; this spec covers the complete environment instead (statement 2553-2563, proof span(s) [(2565, 2623)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Editorial note. The statement contains an authorial (3)/(2) numbering slip; the argument as written is complete and unaffected.

Source reference. Original definition 0DBD: sources/stacks-chow/tags.txt, source lines 16693–16693. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0202: sources/stacks-chow/tags.txt, source lines 2619–2619. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01R5: sources/stacks-chow/tags.txt, source lines 2290–2290. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let T be an affine scheme. Each of the following types of families of maps with target T has a refinement by a standard ph covering:*

(1) *any Zariski open covering of T,*
(2) *$\{W_{ji} \to T\}_{j=1,\ldots,m, i=1,\ldots n_j}$ where $\{W_{ji} \to U_j\}_{i=1,\ldots,n_j}$ and $\{U_j \to T\}_{j=1,\ldots,m}$ are standard ph coverings.*

Proof. Part (1) follows from the fact that any Zariski open covering of T can be refined by a finite affine open covering.

Proof of (3). Choose $U \to T$ proper surjective and $U = \bigcup_{j=1,\ldots,m} U_j$ as in Definition 0DBD. Choose $W_j \to U_j$ proper surjective and $W_j = \bigcup W_{ji}$ as in Definition 0DBD. By Chow's lemma (Limits, Lemma 0202) we can find $W'_j \to W_j$ proper surjective and closed immersions $W'_j \to \mathbf{P}^{e_j}_{U_j}$. Thus, after replacing W_j by W'_j and $W_j = \bigcup W_{ji}$ by a suitable affine open covering of W'_j, we may assume there is a closed immersion $W_j \subset \mathbf{P}^{e_j}_{U_j}$ for all $j = 1, \ldots, m$.

Let $\overline{W}_j \subset \mathbf{P}^{e_j}_U$ be the scheme theoretic closure of W_j. Then $W_j \subset \overline{W}_j$ is an open subscheme; in fact W_j is the inverse image of $U_j \subset U$ under the morphism $\overline{W}_j \to U$. (To see this use that $W_j \to \mathbf{P}^{e_j}_U$ is quasi-compact and hence formation of the scheme

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theoretic image commutes with restriction to opens, see Morphisms, Section 01R5.) Let $Z_j = U \setminus U_j$ with reduced induced closed subscheme structure. Then

$$V_j = \overline{W}_j \amalg Z_j \to U$$

is proper surjective and the open subscheme $W_j \subset V_j$ is the inverse image of U_j. Hence for $v \in V_j$, $v \notin W_j$ we can pick an affine open neighbourhood $v \in V_{j,v} \subset V_j$ which maps into $U_{j'}$ for some $1 \leq j' \leq m$.

To finish the proof we consider the proper surjective morphism

$$V = V_1 \times_U V_2 \times_U \ldots \times_U V_m \longrightarrow U \longrightarrow T$$

and the covering of V by the affine opens

$$V_{1,v_1} \times_U \ldots \times_U V_{j-1,v_{j-1}} \times_U W_{ji} \times_U V_{j+1,v_{j+1}} \times_U \ldots \times_U V_{m,v_m}$$

These do indeed form a covering, because each point of U is in some U_j and the inverse image of U_j in V is equal to $V_1 \times \ldots \times V_{j-1} \times W_j \times V_{j+1} \times \ldots \times V_m$. Observe that the morphism from the affine open displayed above to T factors through W_{ji} thus we obtain a refinement. Finally, we only need a finite number of these affine opens as V is quasi-compact (as a scheme proper over the affine scheme T). $\square$